



RECOVERY REFERENCE

VIRTUAL CONTROL
VMWARE CLOUD FOUNDATION SOLUTIONS

VCF Appliance Filesystem (fsck) Recovery & GRUB Access

Per-component boot-repair procedures, filesystem layouts, and GRUB credentials for the VCF 9 management domain.

FSCK RECOVERY

GRUB

NSX MANAGER

PHOTON APPLIANCES

BOOT REPAIR

VCF 9.0

VMware Cloud Foundation

VCF Appliance Filesystem (fsck) Recovery & GRUB Access Reference

Prepared by: Virtual Control LLC **Date:** May 29, 2026 **Document Version:** 1.0 **Classification:** Internal — VCF Management Domain **Applies to:** VMware Cloud Foundation 9.0 (NSX 9.0.x, VCF Operations 9.0.x, vCenter 9.0.x)

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1. Purpose and Scope

This reference documents the procedure for repairing a corrupted root filesystem on a VMware Cloud Foundation 9.0 appliance that fails to boot, and records the **material differences between the two appliance families** in the management domain — because the boot-repair path is *not* identical across them.

The single most common trigger for this condition is an **ungraceful shutdown** (power loss, host hard-off, or a management VM being killed mid-write). The root filesystem is left with metadata inconsistencies that the automatic boot-time `fsck` (run in safe "preen" mode) will not repair unattended. The appliance then either drops to an emergency shell or enters a reboot loop, and never reaches the network.

The repair is a one-time, manual `fsck` invoked through the GRUB bootloader. The two families differ in:

- the **symptom** presented at the console,
- whether the **GRUB menu is password-protected**, and
- the **root device path** used for a manual `fsck`.

Get those three right per family and the recovery is routine.

2. When This Applies — Recognizing the Symptom

This procedure applies when an appliance will not finish booting and the console shows a filesystem check failure. Match the console output to one of the two patterns below.

2.1 Photon-Family Symptom (dracut emergency)

Photon OS-based appliances drop into the **dracut emergency shell**:

```
[FAILED] Failed to start File System Check on /dev/disk/by-uuid/<UUID>
[DEPEND] Dependency failed for /sysroot
[DEPEND] Dependency failed for Initrd Root File System
You are in emergency mode. ... "Press Enter for maintenance" / Control-D to continue
```

Pressing Enter may loop back without producing a usable shell if systemd is too damaged to spawn a login.

2.2 NSX-Family Symptom (panic reboot loop)

NSX (Ubuntu-based) appliances do **not** stop at a maintenance prompt. The initramfs runs `fsck -a`, fails, and the kernel's `panic=` argument forces an automatic reboot — producing an endless loop:

```
/dev/sda2 contains a file system with errors, check forced.
Inodes that were part of a corrupted orphan linked list found.
/dev/sda2: UNEXPECTED INCONSISTENCY; RUN fsck MANUALLY.
      (i.e., without -a or -p options)
fsck exited with status code 4
Failure: File system check of the root filesystem failed
The root filesystem on /dev/sda2 requires a manual fsck
Rebooting automatically due to panic= boot argument
```

Note: the reboot loop is not an obstacle — the GRUB menu re-appears on every loop, giving repeated opportunities to interrupt the boot and apply the repair.

3. Appliance Families and Filesystem Layout

Attribute	Photon Family	NSX Family
Members	SDDC Manager, VCF Operations, VCF Operations for Logs, VCF Operations Collector, Fleet / LCM, vCenter (VCSA)	NSX Manager, NSX Edge
Guest OS	VMware Photon OS	Ubuntu (kernel <code>5.15.x-nnN-server</code>)
Boot failure mode	dracut emergency shell / <code>/sysroot</code> mount failure	<code>fsck -a</code> failure → <code>panic=</code> auto-reboot loop
Root device (for manual fsck)	<code>/dev/mapper/vg_root_0-lv_root_0</code> (LVM)	<code>/dev/sda2</code> (ext4, direct partition)
GRUB menu password-protected	No (per observed procedure)	Yes — username + password required to edit an entry

dm-verity caveat on manual fsck	Possible — if the root LV is verity-sealed, a manual fsck breaks the signature and the appliance will not boot. Verify verity state first.	Not applicable to <code>/dev/sda2</code> (it is fsck'd at every boot, so it is not verity-sealed)
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Verify before acting on vCenter (VCSA): it is grouped with the Photon family above, but confirm the GRUB protection state and root device path on the specific build before performing a manual `fsck` — VCSA hardening has varied across versions.

4. GRUB Access and Credentials

4.1 Reaching the GRUB Menu

On a healthy boot the GRUB menu is hidden. To force it:

- Reboot the appliance (or, for an NSX appliance already in a panic loop, simply watch the console — GRUB appears each cycle).
- The instant the menu appears, press an **arrow key** (or **left SHIFT / ESC**) to halt the countdown.
- Highlight the default entry (the `Ubuntu` / Photon entry) and press `e` to edit.

4.2 GRUB Credentials by Family

Family	Username	Password	Notes
NSX Manager / NSX Edge	root	NSX@VM!waR10	This is the default GRUB password and is distinct from the appliance OS root password . Required before any boot entry can be edited.
Photon family (SDDC Mgr, VCF Ops, Logs, Collector, Fleet/LCM)	No GRUB credential prompt observed — pressing <code>e</code> opens the editor directly.		If a build does prompt, the GRUB superuser is <code>root</code> ; obtain the password from the VCF Operations Locker.

If NSX@VM!waR10 is rejected, the GRUB password was changed at deployment (non-default). Do not retry blindly — reset it first using Broadcom KB **417635** (*Resetting GRUB password on NSX Manager Appliance*), then return to this procedure.

OS root passwords (needed in Method 2 if the emergency shell prompts) are set per appliance at deploy time and are retrievable from the **VCF Operations Locker**. There is no single shared default.

5. Method 1 — Boot With `fsck.repair=yes` (Preferred)

This is the Broadcom-documented first response (KB **326323**). It instructs the boot-time `fsck` to run in repair mode (answer "yes" to all repairs) for this one boot only. The edit is **not persisted** — a normal power-cycle reverts it.

1. Reach the GRUB editor (Section 4.1) and authenticate if required (Section 4.2).
2. Locate the line beginning with `linux` (the kernel line — it contains `root=UUID=...` and, on NSX, `panic=...`).
3. This line **wraps across several display rows**; the trailing `\` on each row is word-wrap, not content, and a `↓` indicator means it continues below the visible area. Move the cursor into the line and press `Ctrl+E` (End) to jump to its true end. (`Ctrl+A` returns to the start if you overshoot.)
4. Append a single space, then exactly:

```
fsck.repair=yes
```

5. Press `Ctrl-X` (or **F10**) to boot.
6. A **blinking cursor with no output is expected** — `fsck` is repairing the root filesystem. Allow **5–20 minutes**. **Do not interrupt it.**
7. Outcome: the appliance boots through to its login banner, or it reboots itself once and then comes up clean. If it returns to the same failure, proceed to Method 2.

Result on a healthy repair: the appliance reaches the network and its login banner. Always follow with a management-plane health check (e.g. `NSX GET /api/v1/cluster/status`) before declaring recovery — a successful boot is necessary but not sufficient.

6. Method 2 — Emergency Shell + Manual fsck

Use only if Method 1 returns to the same failure.

1. Reach the GRUB editor and authenticate (Section 4.2).
2. On the `linux` line: - **NSX family:** remove the `panic=...` token (so a `fsck` prompt cannot trigger an auto-reboot), then append `systemd.unit=emergency.target`. - **Photon family:** append `systemd.unit=emergency.target` (or `init=/bin/bash`).
3. Press `Ctrl-X` to boot. Provide the OS root password if prompted (VCF Operations Locker).
4. Run the manual repair against the correct root device for the family:

```
# NSX family:
e2fsck -y /dev/sda2

# Photon family:
fsck -y /dev/mapper/vg_root_0-lv_root_0
```

5. **Re-run until it reports clean.** A badly damaged ext4 root commonly needs **2–3 passes** before no errors remain.
6. Reboot:

```
reboot -f
```

dm-verity (Photon family only): if the root logical volume is dm-verity-sealed, a manual `fsck` invalidates the verity signature and the appliance will not boot afterward. Confirm the verity state before running `fsck`

manually on a Photon root LV. This caveat does not apply to the NSX `/dev/sda2` root.

7. Escalation — Redeploy via Fleet Management

If both fsck methods fail to produce a clean, bootable filesystem, replace the component.

- Path: **VCF Operations → Fleet Management → Lifecycle → VCF Management → Components → delete the failed component → Add Component → redeploy.**
- SDDC Manager is **deprecated for this workflow in VCF 9** — use Fleet Management.
- Redeploy replaces only the failed component; the remaining appliances are preserved.

Backup prerequisite for NSX: the VCF "Redeploy First Node of Failed NSX Manager Cluster" workflow requires a matching backup file. Without a current NSX backup, redeploy means total loss of NSX configuration (segments, T0/T1, NAT, DFW, edges). **Always exhaust the fsck repair before redeploying NSX.**

8. Operational Cautions

- **Do not power-cycle an appliance mid- fsck .** Interrupting a repair (or interrupting Corfu/database replay on NSX) is itself a mid-write reset and can deepen the corruption.
- **The GRUB edit is one boot only.** It is never saved to disk; a clean power-cycle reverts it. There is no cleanup step and no risk of leaving a persistent boot change.
- **A clean boot is not proof of health.** Verify the management plane after recovery. NSX in particular can answer on port 443 while the manager role is still down (see the Error 101 / Corfu condition, KB 390592).
- **One ungraceful shutdown can damage several appliances.** After a power event, expect to repeat this procedure on more than one VM. Check each console rather than assuming the first recovery was isolated.
- **Single-quote passwords** in any shell command to avoid history-expansion on special characters.

9. Verified Recovery History

Date	Appliance	Family	Condition	Resolution
2026-04-23	VCF Operations for Logs 9.0.1	Photon	dracut emergency / <code>/sysroot</code> mount failure	Single <code>fsck.repair=yes</code> boot (Method 1) — booted clean to login.
2026-05-05 → 2026-05-13	SDDC Manager	Photon	Filesystem damage following power outage	<code>fsck.repair=yes</code> (Method 1) recovered it; flagged as highest-risk for recurrence on subsequent power cycles.
2026-05-29	NSX Manager (192.168.1.71)	NSX	Root <code>/dev/sda2</code> (ext4) corruption after ungraceful shutdown; <code>panic=60</code> reboot loop; UNEXPECTED INCONSISTENCY; RUN <code>fsck</code> MANUALLY	GRUB authenticated with <code>root / NSX@VM!WaR10</code> ; <code>fsck.repair=yes</code> appended to the <code>linux</code> line (Method 1).

10. Document Information

Field	Detail
Title	VCF Appliance Filesystem (fsck) Recovery & GRUB Access Reference
Author	Virtual Control LLC
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Classification	Internal — VCF Management Domain
Applies to	VMware Cloud Foundation 9.0 (NSX 9.0.x, VCF Operations 9.0.x, vCenter 9.0.x)
Primary references	Broadcom KB 326323 (boot-time fsck recovery); KB 417635 (NSX GRUB password reset); KB 426832 (NSX root/GRUB password recovery); KB 390592 (NSX Error 101 / Corfu)
Related documents	NSX Manager 9.0.1 Cold Start Service Failure — RCA & Recovery

Revision History

Version	Date	Author	Summary
1.0	2026-05-29	Virtual Control LLC	Initial release. Documents the two appliance-family symptoms and filesystem layouts, GRUB access and credentials (including the NSX default GRUB credential), Method 1 (<code>fsck.repair=yes</code>) and Method 2 (emergency shell + manual fsck) procedures, redeploy escalation, and verified recovery history.